

Additive Progress on the Alaoglu–Erdős Conjecture

Kummer–Chebotarev incompatibility, mantissa-orbit entropy,
and a rigidity barrier for auxiliary polynomials

Research working note prepared for Vladimir Reshetnikov

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Abstract

Let $x \in \mathbb{R}$ and suppose that $m = 2^x$ and $n = 3^x$ are integers. The Alaoglu–Erdős conjecture asserts that x must be an integer. This companion report is deliberately additive: it does not repeat the large survey and proof search in Report A. It develops three pieces of new infrastructure.

First, it associates to $(2, 3, m, n)$ an exact order- ℓ Kummer compatibility test. Kummer theory and Chebotarev then give closed formulas for the density of primes at which a common residue exponent exists. For a hypothetical nonintegral counterexample that density is asymptotic to $1/\ell$, while incompatibility has density tending to one. The rank-three case, which initially appears exceptional, is shown to be transverse; its exact compatibility density is $\ell^{-1} - \ell^{-2} + \ell^{-3}$.

Second, the normalized rational points

$$U_j = 2^{\{jx\}} = \frac{m^j}{2^{\lfloor jx \rfloor}}, \quad V_j = 3^{\{jx\}} = \frac{n^j}{3^{\lfloor jx \rfloor}}$$

are studied simultaneously in the real and finite topologies. Conditional on any real subarc of the exponential curve, their order- ℓ residue symbols are uniformly distributed on the image of an explicit 2×2 matrix. At the Chebotarev-generic primes that matrix is invertible, so the finite residue data has full two-dimensional entropy even on an arbitrarily short real arc. This is a rigorous archimedean-to-finite-place theorem, but it points in the opposite direction from a naive local-exponent proof.

Third, an exact local zero estimate is proved. If an algebraic-coefficient polynomial is restricted to $z \mapsto (2^z, 3^z)$ at an algebraic point of the curve, its order of contact is exactly its ordinary bivariate multiplicity. Thus the roughly D^2 coefficients of a degree- D auxiliary polynomial cannot buy order- D^2 directional vanishing; only order $O(D)$ is possible. This identifies, rather than merely gestures at, the obstruction in a broad class of interpolation-determinant attacks.

The conjecture is not claimed solved here. The report ends with three precise bridge lemmas which would finish it, a proof-status ledger, a finite-field verification program, and a staged Lean formalization plan.

Contents

1	Scope, novelty boundary, and proof status	4
1.1	The statement	4
1.2	How this document avoids duplicating Report A	4
1.3	What is proved, and what is not	4

2	Baseline reductions and the multiplicative-rank floor	5
2.1	Rational and algebraic exponents are already excluded	5
2.2	Multiplicative rank	6
2.3	The rank-three normal form and its transversality	6
3	Kummer residue coordinates	7
3.1	The Kummer space	7
3.2	Residue-symbol coordinates	7
3.3	Exact compatibility versus a vanishing determinant	8
4	Exact Kummer–Chebotarev density laws	8
4.1	The universal density formula	8
4.2	Rank four	9
4.3	Rank three: the general hyperplane count	9
4.4	Rank three for a hypothetical counterexample	10
4.5	The projective determinant law	11
4.6	Why this is an obstruction, not yet a contradiction	11
5	The normalized mantissa orbit	12
5.1	A dense rational orbit on a compact arc	12
5.2	Joint equidistribution of the index and the carry	12
5.3	Residue symbols of the mantissa orbit	13
5.4	Interpretation: a rigorous anti-continuity result	13
6	A rigidity barrier for auxiliary polynomials	14
6.1	Exact equality of directional order and ordinary multiplicity	14
6.2	Why the tempting dimension count fails	15
6.3	A Wronskian companion	15
7	Near collisions and a selected-prime bridge	16
7.1	Simultaneous near-power differences	16
7.2	Common divisors are Kummer-compatible	16
7.3	Why density alone is insufficient	17
7.4	A precise sparse-support target	17
8	The one-more-base route	18
9	Three proof bridges and one no-go theorem	19
9.1	Bridge I: universal local compatibility	19
9.2	Bridge II: selected-prime Frobenius escape	19
9.3	Bridge III: one more algebraic power	20
9.4	No-go theorem: exact high directional contact	20
9.5	Architecture scorecard	20

10 Finite verification and reproducible experiments	21
10.1 Exhaustive checker	21
10.2 Arithmetic experiments suggested by the theory	22
11 Lean formalization plan	23
11.1 Layer 0: finite combinatorics	23
11.2 Layer 1: valuation lattices and rank	23
11.3 Layer 2: irrational rotation and mantissa entropy	23
11.4 Layer 3: Kummer theory and Chebotarev	24
11.5 Layer 4: analytic local rigidity	24
11.6 Recommended theorem order	24
12 Research agenda after this report	25
12.1 Highest-priority questions	25
12.2 Directions now deprioritized	25
12.3 A focused next experiment	26
13 Proof-status ledger and conclusion	26
A A compact derivation of the rank-three compatibility formula	27
B Saturation index from a prime-exponent matrix	27
C Bibliographic notes	28

1 Scope, novelty boundary, and proof status

1.1 The statement

We write the conjecture in the following form.

Conjecture 1.1 (Alaoglu–Erdős). *If $x \in \mathbb{R}$ and $2^x, 3^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

Only positive integers can occur: if $x < 0$, then $0 < 2^x < 1$, while if $x = 0$ the conclusion is immediate. Hence every putative counterexample has $x > 0$ and $m = 2^x, n = 3^x \in \mathbb{Z}_{>0}$.

The external reports about recent machine-assisted mathematical discoveries are treated as motivation, not as evidence. Every assertion used below is proved in the text or reduced to a named, standard theorem. In particular, no unpublished announcement is used as a lemma.

1.2 How this document avoids duplicating Report A

The attached source corpus was decompressed and used as an exclusion map. Its machine-audited metadata is recorded here so that the relationship between the documents is reproducible:

Source file	<code>alaoglu-erdos-unified-report-A.tex.gz</code>
Decompressed lines	20,925
Whitespace-delimited words	113,833
Structural headings	491

The following exact strings were also checked against Report A. This is only a lexical guard against accidental repetition, not a claim of priority in the literature.

String identifying a result developed below	Present verbatim in Report A?
<code>mantissa-orbit entropy</code>	no
<code>transcendental-direction rigidity</code>	no
<code>fixed-t Kummer density</code>	no
<code>q^3-q+1</code>	no
<code>rank-three transversality</code>	no

The baseline reductions in section 2 are included only to make the new arguments self-contained. The substantive additions begin with the rank-three transversality lemma, the exact Kummer counts, the conditional mantissa equidistribution theorem, and the auxiliary-polynomial rigidity theorem.

1.3 What is proved, and what is not

The main unconditional conclusions of this report are as follows.

1. Every nonintegral counterexample has multiplicative rank at least three: $\text{rank}(2, 3, m, n) \geq 3$.
2. In rank three, the unique multiplicative relation is necessarily transverse between the input pair $(2, 3)$ and output pair (m, n) .
3. For every sufficiently large odd prime ℓ , the relative Chebotarev density of primes admitting a common order- ℓ residue exponent is given by an exact formula. It is $\ell^{-1} - \ell^{-2} + \ell^{-3}$ in rank three and $\ell^{-1} - \ell^{-3} + \ell^{-4}$ in rank four.

4. At a density $1 - O(1/\ell)$ of primes, the normalized mantissa orbit has all ℓ^2 residue-symbol pairs, uniformly, even after conditioning on an arbitrarily short real interval.
5. Algebraic auxiliary polynomials have no anomalously high directional contact with the curve $z \mapsto (2^z, 3^z)$ at its algebraic points.
6. A common prime divisor of the simultaneous near-power differences $m^b - 2^a$ and $n^b - 3^a$ is forced into the rare Kummer-compatible Chebotarev set whenever $\ell \nmid b$.

None of these statements by itself proves conjecture 1.1. The precise remaining gaps are isolated in section 9. The most important negative conclusion is also constructive: simply transporting the real exponent x to a p -adic or finite-field exponent cannot work at generic primes, because a hypothetical counterexample is provably incompatible at almost all of them.

2 Baseline reductions and the multiplicative-rank floor

Throughout the rest of the report, a *counterexample datum* means a triple

$$(x, m, n), \quad x \in \mathbb{R} \setminus \mathbb{Z}, \quad m = 2^x \in \mathbb{Z}_{>0}, \quad n = 3^x \in \mathbb{Z}_{>0}.$$

The point of introducing the term is not to assume that such a datum exists, but to make conditional deductions compact.

2.1 Rational and algebraic exponents are already excluded

Proposition 2.1. *If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

Proof. Write $x = a/b$ with $a \in \mathbb{Z}$, $b \in \mathbb{Z}_{>0}$, and $\gcd(a, b) = 1$. Since $2^{a/b}$ is a positive integer m , one has $2^a = m^b$. Comparing the 2-adic valuations gives $a = bv_2(m)$. Hence $b \mid a$, so $b = 1$. $\square$

Corollary 2.2. *In a counterexample datum, x is transcendental.*

Proof. By proposition 2.1, x is irrational. If it were algebraic, then the Gelfond–Schneider theorem, applied to the algebraic base 2 and the algebraic irrational exponent x , would make every value of 2^x transcendental. This contradicts $2^x = m \in \mathbb{Z}$. $\square$

Set

$$\alpha = \log_2 3.$$

Proposition 2.3. *The number α is transcendental.*

Proof. It is not rational, since $\alpha = r/s \in \mathbb{Q}$ would imply $2^r = 3^s$ after clearing denominators. If α were algebraic irrational, then Gelfond–Schneider would make 2^α transcendental, whereas $2^\alpha = 3$. $\square$

The conjecture can therefore be viewed as a very special rank-one relation between a 2×2 matrix of logarithms of positive integers:

$$\det \begin{pmatrix} \log 2 & \log 3 \\ \log m & \log n \end{pmatrix} = 0, \quad (\log m, \log n) = x(\log 2, \log 3). \quad (1)$$

The Four Exponentials Conjecture would rule out this matrix when x is irrational. The integrality of m, n is the extra datum one hopes to exploit.

2.2 Multiplicative rank

For a finitely generated subgroup $\Gamma \subset \mathbb{Q}_{>0}^\times$, its rank means its rank as a free abelian group. Put

$$\Gamma = \langle 2, 3, m, n \rangle \subset \mathbb{Q}_{>0}^\times, \quad r = \text{rank } \Gamma.$$

Since 2 and 3 are multiplicatively independent, $r \geq 2$.

Proposition 2.4 (rank floor). *For a counterexample datum one has $r \geq 3$.*

Proof. Suppose $r = 2$. Since the valuation vectors of 2 and 3 are independent, any prime outside $\{2, 3\}$ occurring in m or n would raise the rank. Consequently

$$m = 2^a 3^b, \quad n = 2^c 3^d$$

with $a, b, c, d \in \mathbb{Z}_{\geq 0}$. Dividing logarithms by $\log 2$ gives

$$x = a + b\alpha, \quad x\alpha = c + d\alpha.$$

Eliminating x yields

$$b\alpha^2 + (a - d)\alpha - c = 0.$$

By proposition 2.3, all three rational coefficients vanish. Thus $b = c = 0$ and $a = d$, whence $x = a \in \mathbb{Z}$, contrary to the definition of a counterexample datum. $\square$

Corollary 2.5. *In a counterexample datum, at least one of m, n has a prime divisor outside $\{2, 3\}$.*

Proof. If both were $\{2, 3\}$ -smooth, then Γ would have rank two, contrary to proposition 2.4. $\square$

2.3 The rank-three normal form and its transversality

If $r = 3$, the homomorphism $\mathbb{Z}^4 \rightarrow \Gamma$ sending the standard basis to $2, 3, m, n$ has a rank-one kernel. Let

$$2^A 3^B m^C n^D = 1, \quad (A, B, C, D) \in \mathbb{Z}^4 \tag{2}$$

be a primitive generator of that kernel, unique up to sign. Substitution of $m = 2^x, n = 3^x$ gives

$$(A + Cx) + (B + Dx)\alpha = 0, \quad x = -\frac{A + B\alpha}{C + D\alpha}. \tag{3}$$

The denominator is nonzero: otherwise $C + D\alpha = 0$, and the transcendence of α would force $C = D = 0$, which would make $2^A 3^B = 1$ and hence $A = B = 0$.

Lemma 2.6 (rank-three transversality). *For a rank-three counterexample datum, the primitive relation (2) satisfies*

$$AD - BC \neq 0.$$

Equivalently, the coefficient vectors (A, B) and (C, D) are not proportional over $\mathbb{Q}$.

Proof. If $(A, B) = \lambda(C, D)$ for some $\lambda \in \mathbb{Q}$, then (3) becomes

$$(x + \lambda)(C + D\alpha) = 0.$$

The second factor is nonzero, so $x = -\lambda \in \mathbb{Q}$. By proposition 2.1, $x \in \mathbb{Z}$, a contradiction. $\square$

This tiny determinant will be responsible for the clean rank-three Chebotarev formula. It removes an apparent exceptional case that would otherwise double the compatible density.

3 Kummer residue coordinates

3.1 The Kummer space

Fix an odd prime ℓ and put

$$K_\ell = \mathbb{Q}(\zeta_\ell),$$

where ζ_ℓ is a chosen primitive ℓ th root of unity. Let V_ℓ be the $\mathbb{F}_\ell$ -subspace of $K_\ell^\times / K_\ell^{\times\ell}$ generated by the classes of $2, 3, m, n$, and write

$$r_\ell = \dim_{\mathbb{F}_\ell} V_\ell.$$

Kummer theory identifies

$$\text{Gal}(K_\ell(2^{1/\ell}, 3^{1/\ell}, m^{1/\ell}, n^{1/\ell})/K_\ell) \cong \text{Hom}_{\mathbb{F}_\ell}(V_\ell, \mu_\ell), \quad (4)$$

so this Galois group has order ℓ^{r_ℓ} .

The dimension r_ℓ agrees with the ordinary multiplicative rank for all but finitely many ℓ . The finite exceptional set is worth stating correctly; ignoring it is a common source of overstrong Kummer claims.

Lemma 3.1 (eventual Kummer rank). *Let $\Gamma = \langle 2, 3, m, n \rangle$ have rank r . There is a positive integer I_Γ such that, for every odd prime $\ell \nmid I_\Gamma$,*

$$r_\ell = r.$$

Proof. Let S be the finite set of rational primes occurring in $6mn$. The valuation map identifies the group of positive rationals supported on S with the lattice $\mathbb{Z}^S$. The image L of Γ is a rank- r sublattice. Let

$$L^{\text{sat}} = (L \otimes_{\mathbb{Z}} \mathbb{Q}) \cap \mathbb{Z}^S, \quad I_\Gamma = [L^{\text{sat}} : L].$$

For $\ell \nmid I_\Gamma$, reduction modulo ℓ gives an r -dimensional image in $\mathbb{Q}^\times / \mathbb{Q}^{\times\ell}$.

It remains to see that passing from $\mathbb{Q}$ to K_ℓ creates no new ℓ th-power relation among positive rationals. Suppose $a \in \mathbb{Q}_{>0}$ and $a = y^\ell$ for some $y \in K_\ell$. At a rational prime $p \neq \ell$, the cyclotomic extension is unramified, so every valuation of a above p is $v_p(a)$ and must be divisible by ℓ . At $p = \ell$ the ramification index is $\ell - 1$; hence $\ell \mid (\ell - 1)v_\ell(a)$, and coprimality gives $\ell \mid v_\ell(a)$. All rational prime valuations of a are therefore divisible by ℓ , and positivity gives $a \in \mathbb{Q}^{\times\ell}$. Thus the rational Kummer classes inject into the cyclotomic Kummer space, proving the claim. $\square$

We call an odd prime ℓ *good for the datum* if it avoids I_Γ and, in rank three, also avoids the nonzero integer $AD - BC$ from lemma 2.6. Only finitely many odd primes are not good.

3.2 Residue-symbol coordinates

Let

$$L_\ell = K_\ell(2^{1/\ell}, 3^{1/\ell}, m^{1/\ell}, n^{1/\ell}).$$

For a prime ideal $\mathfrak{p}$ of K_ℓ unramified in L_ℓ and not dividing $6mn\ell$, define $\kappa_{\mathfrak{p}}(a) \in \mathbb{F}_\ell$ by

$$\left(\frac{a}{\mathfrak{p}}\right)_\ell = \zeta_\ell^{\kappa_{\mathfrak{p}}(a)}, \quad a \in \{2, 3, m, n\}.$$

Equivalently, κ_p is the Kummer character represented by Frob_p . Put

$$\kappa_p = (u_1, u_2, v_1, v_2) = (\kappa_p(2), \kappa_p(3), \kappa_p(m), \kappa_p(n)) \in \mathbb{F}_\ell^4. \quad (5)$$

Changing the chosen primitive root ζ_ℓ multiplies all four coordinates by one nonzero scalar, so all zero/nonzero and proportionality conditions used below are canonical.

By Chebotarev, the vectors (5) are uniformly distributed in the r_ℓ -dimensional linear subspace

$$W_\ell = V_\ell^\vee \subset \mathbb{F}_\ell^4.$$

More explicitly, every vector in W_ℓ occurs on a set of prime ideals of K_ℓ of relative density ℓ^{-r_ℓ} .

3.3 Exact compatibility versus a vanishing determinant

There are two related local tests, and they must not be conflated.

Definition 3.2 (exact common residue exponent). A vector $(u_1, u_2, v_1, v_2) \in \mathbb{F}_\ell^4$ is *compatible* if there exists $t \in \mathbb{F}_\ell$ such that

$$(v_1, v_2) = t(u_1, u_2).$$

Let $\mathcal{C}_\ell$ denote the set of compatible vectors, and let

$$\mathcal{C}_{\ell,t} = \{(u_1, u_2, tu_1, tu_2) : u_1, u_2 \in \mathbb{F}_\ell\}$$

be the fixed- t compatibility plane.

The projective determinant

$$\Delta(u_1, u_2, v_1, v_2) = u_1 v_2 - u_2 v_1 \quad (6)$$

vanishes on $\mathcal{C}_\ell$, but the converse fails when $(u_1, u_2) = (0, 0)$ and $(v_1, v_2) \neq (0, 0)$. Therefore $\Delta = 0$ is a useful rank detector, not an exact common-exponent detector. The distinction changes the main term beyond first order and is important in formal verification.

Lemma 3.3 (finite compatibility count). *One has*

$$|\mathcal{C}_\ell| = \ell^3 - \ell + 1.$$

Proof. For a nonzero source vector $u = (u_1, u_2)$, the ℓ values of t give ℓ distinct targets tu . This contributes $\ell(\ell^2 - 1)$ vectors. For $u = 0$, exact compatibility permits only the zero target, contributing one more vector. $\square$

Proof-audit warning 3.4. Replacing $|\mathcal{C}_\ell|$ by the number of singular 2×2 matrices, $\ell^3 + \ell^2 - \ell$, silently counts the $\ell^2 - 1$ false positives with zero source and nonzero target. Any Lean formalization should define compatibility by an existential scalar before proving a determinant corollary.

4 Exact Kummer–Chebotarev density laws

4.1 The universal density formula

The Kummer space gives a formula before any rank classification.

Theorem 4.1 (Kummer compatibility density). *For every odd prime ℓ , the relative density among unramified prime ideals $\mathfrak{p}$ of K_ℓ at which a common order- ℓ residue exponent exists is*

$$\text{dens}_\ell(\text{compatible}) = \frac{|W_\ell \cap \mathcal{C}_\ell|}{\ell^{r_\ell}}. \quad (7)$$

For a fixed $t \in \mathbb{F}_\ell$, the corresponding density is

$$\text{dens}_\ell(t) = \frac{|W_\ell \cap \mathcal{C}_{\ell,t}|}{\ell^{r_\ell}}.$$

Proof. The Kummer isomorphism (4) identifies the Galois group with the finite additive space W_ℓ . The Frobenius element at $\mathfrak{p}$ has coordinate vector $\kappa_{\mathfrak{p}}$. Chebotarev assigns density $1/\ell^{r_\ell}$ to each Galois element. Summing over the indicated subsets proves both formulas. $\square$

The content is therefore finite linear algebra. The rank-three transversality lemma makes that linear algebra particularly clean.

4.2 Rank four

Theorem 4.2 (rank-four law). *Suppose $r_\ell = 4$. Then, for every $t \in \mathbb{F}_\ell$,*

$$\text{dens}_\ell(t) = \frac{1}{\ell^2},$$

and the exact compatibility density is

$$\text{dens}_\ell(\text{compatible}) = \frac{\ell^3 - \ell + 1}{\ell^4} = \frac{1}{\ell} - \frac{1}{\ell^3} + \frac{1}{\ell^4}. \quad (8)$$

The incompatibility density is therefore

$$1 - \frac{1}{\ell} + \frac{1}{\ell^3} - \frac{1}{\ell^4}. \quad (9)$$

Proof. Here $W_\ell = \mathbb{F}_\ell^4$. Each fixed- t plane has ℓ^2 points, giving density $\ell^2/\ell^4 = \ell^{-2}$. The union count is lemma 3.3; division by ℓ^4 proves (8). $\square$

The formula can also be seen as a corrected union of ℓ planes. Any two planes $\mathcal{C}_{\ell,t}$ and $\mathcal{C}_{\ell,s}$ with $t \neq s$ meet only at the zero vector. Thus

$$|\mathcal{C}_\ell| = 1 + \ell(\ell^2 - 1).$$

The common zero Frobenius is the only overlap.

4.3 Rank three: the general hyperplane count

Before using transversality, it is useful to record the exact count for an arbitrary hyperplane. Let

$$W = \{(u_1, u_2, v_1, v_2) : Au_1 + Bu_2 + Cv_1 + Dv_2 = 0\} \quad (10)$$

be a hyperplane in $\mathbb{F}_\ell^4$. Define linear forms on $u = (u_1, u_2)$ by

$$L(u) = Au_1 + Bu_2, \quad M(u) = Cu_1 + Du_2.$$

On the fixed- t compatibility plane $v = tu$, the hyperplane equation becomes

$$(L + tM)(u) = 0. \quad (11)$$

Proposition 4.3 (exact hyperplane count). *For the hyperplane (10),*

$$|W \cap \mathcal{C}_\ell| = \begin{cases} 2\ell^2 - 2\ell + 1, & M \neq 0 \text{ and } L \in \mathbb{F}_\ell M, \\ \ell^2 - \ell + 1, & \text{otherwise.} \end{cases}$$

Proof. The zero source contributes the single zero vector. For a nonzero source u , the number of scalars t solving $L(u) + tM(u) = 0$ is one if $M(u) \neq 0$, is ℓ if $L(u) = M(u) = 0$, and is zero if $M(u) = 0 \neq L(u)$.

If $M \neq 0$ and L is proportional to M , their kernels coincide and contain $\ell - 1$ nonzero vectors. There are $\ell^2 - \ell$ nonzero vectors outside that kernel, so the total is

$$1 + (\ell^2 - \ell) + \ell(\ell - 1) = 2\ell^2 - 2\ell + 1.$$

If $M \neq 0$ and L, M are independent, the common kernel is zero and M has $\ell - 1$ nonzero zeros. Hence the total is $1 + (\ell^2 - 1) - (\ell - 1) = \ell^2 - \ell + 1$. Finally, if $M = 0$, then $L \neq 0$; only the $\ell - 1$ nonzero vectors in $\ker L$ work, each with all ℓ values of t , again giving $1 + \ell(\ell - 1)$. $\square$

The larger first case is exactly the nontransverse case. It cannot persist for a nonintegral rank-three counterexample.

4.4 Rank three for a hypothetical counterexample

Theorem 4.4 (transverse rank-three law). *Let (x, m, n) be a rank-three counterexample datum, and let ℓ be good for the datum. Then, for every $t \in \mathbb{F}_\ell$,*

$$\text{dens}_\ell(t) = \frac{1}{\ell^2},$$

and the exact compatibility density is

$$\text{dens}_\ell(\text{compatible}) = \frac{\ell^2 - \ell + 1}{\ell^3} = \frac{1}{\ell} - \frac{1}{\ell^2} + \frac{1}{\ell^3}. \quad (12)$$

Thus incompatibility has density

$$1 - \frac{1}{\ell} + \frac{1}{\ell^2} - \frac{1}{\ell^3}. \quad (13)$$

Proof. For a good ℓ , $r_\ell = 3$ and W_ℓ is the reduction modulo ℓ of the primitive relation (2):

$$Au_1 + Bu_2 + Cv_1 + Dv_2 = 0.$$

Goodness includes $\ell \nmid AD - BC$, so L and M are independent. For each fixed t , the form $L + tM$ is nonzero and has a one-dimensional kernel, so $|W_\ell \cap \mathcal{C}_{\ell,t}| = \ell$. Since $|W_\ell| = \ell^3$, the fixed- t density is ℓ^{-2} .

For distinct s, t , the sets $W_\ell \cap \mathcal{C}_{\ell,s}$ and $W_\ell \cap \mathcal{C}_{\ell,t}$ meet only at zero. Their union therefore has

$$1 + \ell(\ell - 1) = \ell^2 - \ell + 1$$

points, proving (12). $\square$

Remark 4.5 (fixed exponents are uniformly expensive). The fixed- t density is exactly ℓ^{-2} in both ranks three and four. Rank changes only the density of the completely split zero vector and hence the overlap correction when the ℓ possible exponents are united. This makes the main term $1/\ell$ transparent: there are ℓ candidate exponents, each imposing two independent residue equations.

4.5 The projective determinant law

The determinant Δ from (6) is not the exact detector, but it will control the finite entropy of the mantissa orbit. Its density is also elementary.

Proposition 4.6 (projective rank law). *Let ℓ be an odd prime.*

1. *If $r_\ell = 4$, then*

$$\text{dens}_\ell(\Delta = 0) = \frac{\ell^3 + \ell^2 - \ell}{\ell^4} = \frac{1}{\ell} + \frac{1}{\ell^2} - \frac{1}{\ell^3}.$$

2. *If (x, m, n) is a rank-three counterexample datum and ℓ is good, then*

$$\text{dens}_\ell(\Delta = 0) = \frac{1}{\ell}.$$

Proof. In rank four, $\Delta = 0$ means that a 2×2 matrix has rank at most one. There is one zero matrix and

$$\frac{(\ell^2 - 1)^2}{\ell - 1}$$

rank-one matrices, for a total of $\ell^3 + \ell^2 - \ell$.

In rank three, W_ℓ is the hyperplane whose coefficient matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ is invertible modulo ℓ . Left and right changes of basis preserve singularity and reduce the hyperplane to $a + d = 0$. A matrix $\begin{pmatrix} a & b \\ c & -a \end{pmatrix}$ is singular when $a^2 + bc = 0$. If $b \neq 0$, there are $\ell(\ell - 1)$ choices, and if $b = 0$ one must have $a = 0$ while c is arbitrary, giving another ℓ choices. The total is ℓ^2 out of ℓ^3 . $\square$

Corollary 4.7 (generic local incompatibility). *For every hypothetical counterexample datum and every sufficiently large odd prime ℓ , exact common-exponent compatibility occurs on a relative Chebotarev set of density $1/\ell + O(1/\ell^2)$, while projective rank collapse occurs on a set of density $1/\ell + O(1/\ell^2)$. In particular, both kinds of local alignment fail on a density tending to one as $\ell \rightarrow \infty$.*

Proof. By proposition 2.4, the rank is three or four. Apply theorems 4.2 and 4.4 and proposition 4.6 outside the finite set of bad primes. $\square$

4.6 Why this is an obstruction, not yet a contradiction

The real equation $(\log m, \log n) = x(\log 2, \log 3)$ does not define an element $x \bmod \ell$ and does not imply residue compatibility. In fact, corollary 4.7 says that if a counterexample exists, then an attempted common residue exponent fails at almost every Kummer prime as ℓ grows. Hence a proof cannot proceed by the sentence “reduce the real exponent modulo ℓ ”; there is no such reduction.

A successful local proof would need one of the following genuinely new bridges:

- a reason to select primes from the sparse compatible Chebotarev set;
- a global product-formula argument that couples many incompatible places without assigning a local exponent;
- or a theorem that turns the integrality of the entire orbit, rather than one pair of values, into a finite-place constraint.

The next section tests the third possibility exactly.

5 The normalized mantissa orbit

5.1 A dense rational orbit on a compact arc

For $j \geq 0$, define

$$k_j = \lfloor jx \rfloor, \quad \theta_j = \{jx\}, \quad U_j = \frac{m^j}{2^{k_j}} = 2^{\theta_j}, \quad V_j = \frac{n^j}{3^{k_j}} = 3^{\theta_j}. \quad (14)$$

Thus

$$1 \leq U_j < 2, \quad 1 \leq V_j < 3,$$

and every (U_j, V_j) is a rational point on the compact analytic arc

$$\mathcal{A} = \{(2^t, 3^t) : 0 \leq t < 1\}.$$

Since x is irrational, (θ_j) is dense and uniformly distributed in $[0, 1)$; hence the rational points (14) are dense on $\mathcal{A}$.

This reformulation packages all positive points of the additive lattice $\mathbb{Z} + \mathbb{Z}x$. It is stronger than looking at (m, n) alone, but its height also grows exponentially with j , so density by itself does not violate rational-point counting theorems.

5.2 Joint equidistribution of the index and the carry

The pair (j, k_j) is the discrete coding of the irrational rotation by x . The following exact refinement is the key input.

Theorem 5.1 (conditional carry equidistribution). *Let x be irrational and let $q \geq 2$ be an integer. For every residue pair $(r, s) \in (\mathbb{Z}/q\mathbb{Z})^2$ and every interval $I \subset [0, 1)$,*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \#\{0 \leq j < N : j \equiv r \pmod{q}, \lfloor jx \rfloor \equiv s \pmod{q}, \{jx\} \in I\} = \frac{|I|}{q^2}. \quad (15)$$

The same holds for finite unions of intervals whose boundaries have measure zero. Equivalently,

$$(\{jx\}, j \bmod q, \lfloor jx \rfloor \bmod q)$$

is equidistributed for the product of Lebesgue measure and uniform counting measure on $(\mathbb{Z}/q\mathbb{Z})^2$.

Proof. Fix r and write $j = r + qh$. Put

$$y_h = \frac{jx}{q} = \frac{rx}{q} + hx, \quad t_h = \{y_h\}.$$

Because x is irrational, (t_h) is uniformly distributed in $[0, 1)$. Now

$$jx = q\lfloor y_h \rfloor + qt_h,$$

so

$$\lfloor jx \rfloor \bmod q = \lfloor qt_h \rfloor, \quad \{jx\} = \{qt_h\}.$$

The conditions $\lfloor jx \rfloor \equiv s \pmod{q}$ and $\{jx\} \in I$ are therefore equivalent to

$$t_h \in \frac{s+I}{q} = \left\{ \frac{s+u}{q} : u \in I \right\},$$

an interval of length $|I|/q$. Among indices with $j \equiv r \pmod{q}$ this has conditional density $|I|/q$; the residue class r itself has density $1/q$. Their product is $|I|/q^2$. $\square$

The theorem is elementary, but it supplies the missing quantitative statement that “real closeness” may be imposed simultaneously with arbitrary residues of both j and its carry $\lfloor jx \rfloor$.

5.3 Residue symbols of the mantissa orbit

Fix an odd prime ℓ and an unramified prime ideal $\mathfrak{p}$ as in section 3. Multiplicativity of the Kummer character and (14) give

$$\begin{pmatrix} \kappa_{\mathfrak{p}}(U_j) \\ \kappa_{\mathfrak{p}}(V_j) \end{pmatrix} = M_{\mathfrak{p}} \begin{pmatrix} j \\ k_j \end{pmatrix} \pmod{\ell}, \quad M_{\mathfrak{p}} = \begin{pmatrix} \kappa_{\mathfrak{p}}(m) & -\kappa_{\mathfrak{p}}(2) \\ \kappa_{\mathfrak{p}}(n) & -\kappa_{\mathfrak{p}}(3) \end{pmatrix}. \quad (16)$$

Its determinant is

$$\det M_{\mathfrak{p}} = \kappa_{\mathfrak{p}}(2)\kappa_{\mathfrak{p}}(n) - \kappa_{\mathfrak{p}}(3)\kappa_{\mathfrak{p}}(m) = \Delta(\kappa_{\mathfrak{p}}). \quad (17)$$

Theorem 5.2 (mantissa-orbit entropy). *Let (x, m, n) be a counterexample datum, let ℓ be an odd prime, and let $\mathfrak{p}$ be an unramified prime ideal as above. For every interval $I \subset [0, 1)$, conditional on $\theta_j \in I$ the vectors*

$$(\kappa_{\mathfrak{p}}(U_j), \kappa_{\mathfrak{p}}(V_j))$$

are uniformly distributed on $\text{im } M_{\mathfrak{p}}$. More precisely, each $w \in \text{im } M_{\mathfrak{p}}$ occurs with absolute natural density

$$\frac{|I|}{|\text{im } M_{\mathfrak{p}}|}.$$

Its conditional density among the indices with $\theta_j \in I$ is therefore $1/|\text{im } M_{\mathfrak{p}}|$. Consequently:

1. *if $\Delta(\kappa_{\mathfrak{p}}) \neq 0$, every one of the ℓ^2 symbol pairs occurs with density $|I|/\ell^2$;*
2. *if $M_{\mathfrak{p}}$ has rank one, every point on its image line occurs with density $|I|/\ell$;*
3. *if $M_{\mathfrak{p}} = 0$, the symbol pair is identically zero.*

Proof. By theorem 5.1, conditional on $\theta_j \in I$, the vector $(j, k_j) \pmod{\ell}$ is uniform on $\mathbb{F}_{\ell}^2$. A linear map sends uniform measure to uniform measure on its image, with every image point having $|\ker M_{\mathfrak{p}}|$ preimages. Equation (16) completes the proof. $\square$

Corollary 5.3 (full local entropy at generic primes). *For a counterexample datum and every sufficiently large good odd prime ℓ , the following holds.*

1. *In multiplicative rank four, a relative Chebotarev proportion*

$$1 - \frac{1}{\ell} - \frac{1}{\ell^2} + \frac{1}{\ell^3}$$

of prime ideals have full mantissa-orbit entropy ℓ^2 on every real subinterval.

2. *In multiplicative rank three, a relative proportion $1 - 1/\ell$ have full entropy.*

Proof. Full entropy is equivalent to $\Delta \neq 0$ by (17). Apply proposition 4.6. $\square$

5.4 Interpretation: a rigorous anti-continuity result

The points (U_j, V_j) can be restricted to an arbitrarily short real arc around any point $(2^t, 3^t)$, yet at a Chebotarev-generic prime their residue symbols still cover all of μ_{ℓ}^2 uniformly. Thus no argument of the following form can be valid:

The real points lie nearly on one tangent direction, so their reductions must have approximately one-dimensional residue data.

There is no finite-field notion of “approximately,” and the exact theorem shows two-dimensional behavior even after arbitrarily severe archimedean conditioning.

This does not make local methods useless. It says that any useful local constraint must involve heights, divisibility by primes that vary with j , or a simultaneous adelic product—not reduction modulo one fixed prime. The selected-prime strategy in section 7 follows precisely that prescription.

6 A rigidity barrier for auxiliary polynomials

Auxiliary-polynomial and interpolation-determinant methods are natural here. A counterexample supplies algebraic points

$$(2^{a+bx}, 3^{a+bx}) = (2^a m^b, 3^a n^b)$$

for all integers a, b for which the displayed quantities are interpreted as rationals, and as integers when $a, b \geq 0$. One might hope to use the roughly D^2 coefficients of a bivariate degree- D polynomial to obtain roughly D^2 orders of vanishing along the one-dimensional exponential curve. The next theorem shows that algebraic coefficients forbid exactly that gain.

6.1 Exact equality of directional order and ordinary multiplicity

For a polynomial $P \in \overline{\mathbb{Q}}[X, Y]$ and a point $(A, B) \in \overline{\mathbb{Q}}^{\times 2}$, let $\text{mult}_{(A,B)} P$ be the least total degree of a nonzero homogeneous term in the Taylor expansion of $P(A + X, B + Y)$. It is zero when $P(A, B) \neq 0$.

Theorem 6.1 (transcendental-direction rigidity). *Let $z_0 \in \mathbb{C}$ be such that*

$$A = 2^{z_0}, \quad B = 3^{z_0}$$

are algebraic and nonzero. For every nonzero $P \in \overline{\mathbb{Q}}[X, Y]$, define

$$F(t) = P(Ae^{t \log 2}, Be^{t \log 3}) = P(2^{z_0+t}, 3^{z_0+t}).$$

Then

$$\text{ord}_{t=0} F = \text{mult}_{(A,B)} P. \tag{18}$$

Proof. Let $d = \text{mult}_{(A,B)} P$ and write

$$P(A + X, B + Y) = H_d(X, Y) + \text{terms of total degree greater than } d,$$

where $H_d \in \overline{\mathbb{Q}}[X, Y]$ is a nonzero homogeneous polynomial of degree d . As $t \rightarrow 0$,

$$Ae^{t \log 2} - A = A(\log 2)t + O(t^2), \quad Be^{t \log 3} - B = B(\log 3)t + O(t^2).$$

Therefore

$$\begin{aligned} F(t) &= H_d(A \log 2, B \log 3) t^d + O(t^{d+1}) \\ &= (\log 2)^d H_d(A, B \alpha) t^d + O(t^{d+1}), \quad \alpha = \log_2 3. \end{aligned}$$

The polynomial $T \mapsto H_d(A, BT)$ is a nonzero polynomial over $\overline{\mathbb{Q}}$: multiplication of its coefficients by the nonzero algebraic numbers A and B cannot annihilate it. By proposition 2.3, α is transcendental over $\mathbb{Q}$ and therefore over $\overline{\mathbb{Q}}$; it does not vanish at $T = \alpha$. Hence the coefficient of t^d is nonzero, proving (18). $\square$

Corollary 6.2 (directional conditions are all mixed conditions). *Under the hypotheses of theorem 6.1, the following are equivalent for an integer $T \geq 1$:*

1. $F^{(j)}(0) = 0$ for $0 \leq j < T$;
2. every mixed partial derivative $\partial_X^r \partial_Y^s P(A, B)$ with $r + s < T$ vanishes;
3. P has ordinary bivariate multiplicity at least T at (A, B) .

Proof. The equivalence of the second and third statements is the definition of multiplicity in characteristic zero. The first is $\text{ord}_0 F \geq T$, which is equivalent to the third by theorem 6.1. $\square$

Corollary 6.3 (degree barrier). *If P has total degree at most D , then*

$$\text{ord}_{t=0} P(2^{z_0+t}, 3^{z_0+t}) \leq D.$$

If P has bidegree at most (D, D) , the order is at most $2D$.

Proof. The multiplicity of a nonzero polynomial at a point cannot exceed its total degree. A polynomial of bidegree (D, D) has total degree at most $2D$. $\square$

The theorem applies at every arithmetic lattice point $z_0 = a + bx$, because $2^{z_0} = 2^a m^b$ and $3^{z_0} = 3^a n^b$ are rational. Hence the obstruction is not confined to the base point $(1, 1)$.

6.2 Why the tempting dimension count fails

The vector space of polynomials of total degree at most D has dimension $\binom{D+2}{2} \asymp D^2/2$. If one viewed $P(2^t, 3^t)$ as an arbitrary one-variable analytic function, it would be tempting to impose almost $D^2/2$ vanishing derivatives at one point. But those derivative equations have coefficients involving powers of $\alpha = \log_2 3$. For algebraic coefficients of P , the transcendence of α separates every coefficient of every power of α . Consequently the first T directional equations are exactly the $\binom{T+1}{2}$ mixed Taylor equations. The apparent one-dimensional problem retains its full two-dimensional arithmetic cost.

This explains a familiar threshold without appealing to vague statements about “not enough parameters.” A successful auxiliary construction must do something genuinely different from exact high contact by an algebraic polynomial. Possibilities include:

- approximate rather than exact contact, together with an unusually strong algebraic-independence measure for α ;
- simultaneous smallness at many points with denominators that cancel in a global determinant;
- an auxiliary object outside $\overline{\mathbb{Q}}[X, Y]$, but with enough arithmetic structure to retain a nonzero lower bound;
- or a finite-place divisibility gain that replaces high archimedean multiplicity.

6.3 A Wronskian companion

For completeness, let $(i_1, j_1), \dots, (i_N, j_N)$ be distinct pairs of nonnegative integers and put

$$f_\nu(z) = 2^{i_\nu z} 3^{j_\nu z}.$$

Their Wronskian is

$$W(f_1, \dots, f_N)(z) = \left(\prod_{\nu=1}^N f_\nu(z) \right) \prod_{1 \leq \mu < \nu \leq N} ((i_\nu - i_\mu) \log 2 + (j_\nu - j_\mu) \log 3), \quad (19)$$

up to the sign determined by the ordering. It never vanishes, since a zero factor would make α rational. Thus these monomials form an extended Chebyshev system on the real line: a nontrivial linear combination with N terms has at most $N - 1$ real zeros counted with multiplicity. This analytic zero estimate is sharp over arbitrary complex coefficients, while theorem 6.1 is the stronger arithmetic local statement for algebraic polynomial coefficients.

Proof-audit warning 6.4 (the auxiliary-polynomial trap). *A claim of order $\gg D$ for $P(2^{z_0+t}, 3^{z_0+t})$ with $P \in \overline{\mathbb{Q}}[X, Y]$ of degree D is false unless $P = 0$. If the proposed construction uses D^2 coefficients to impose D^2 directional derivatives, then either the coefficient field secretly contains α , the equations are only approximate, or the linear system has only the zero solution.*

7 Near collisions and a selected-prime bridge

The Kummer theorem says generic fixed primes are hostile to a local exponent. A more plausible strategy is to let the prime vary with a rational approximation to x . This section identifies exactly where such primes are forced to lie.

7.1 Simultaneous near-power differences

Let $a, b \in \mathbb{Z}_{>0}$ and put

$$\delta = bx - a.$$

Then

$$E_{a,b} = m^b - 2^a = 2^a(2^\delta - 1), \quad F_{a,b} = n^b - 3^a = 3^a(3^\delta - 1). \quad (20)$$

When a/b is a continued-fraction convergent to x , one has $|\delta| \ll 1/b$. For $|\delta| \leq 1$, the mean-value theorem gives constants $c_1, c_2 > 0$, independent of a, b , such that

$$c_1 2^a |\delta| \leq |E_{a,b}| \leq c_2 2^a |\delta|, \quad c_1 3^a |\delta| \leq |F_{a,b}| \leq c_2 3^a |\delta|. \quad (21)$$

Thus the differences are small relatively, but still exponentially large absolutely. Any argument that concludes $E_{a,b} = 0$ merely from $|\delta| \ll 1/b$ has lost an exponential factor.

7.2 Common divisors are Kummer-compatible

Proposition 7.1 (common-divisor localization). *Let ℓ be an odd prime. Let $p \nmid 6mn\ell$ be a rational prime with $p \equiv 1 \pmod{\ell}$, and let $\mathfrak{p}$ be a prime of K_ℓ above p . Suppose*

$$p \mid E_{a,b}, \quad p \mid F_{a,b}, \quad \ell \nmid b.$$

Then $\mathfrak{p}$ is exactly compatible, with common residue exponent

$$t \equiv ab^{-1} \pmod{\ell}.$$

Proof. The divisibilities give

$$m^b \equiv 2^a \pmod{\mathfrak{p}}, \quad n^b \equiv 3^a \pmod{\mathfrak{p}}.$$

Applying the order- ℓ residue character yields

$$b\kappa_{\mathfrak{p}}(m) = a\kappa_{\mathfrak{p}}(2), \quad b\kappa_{\mathfrak{p}}(n) = a\kappa_{\mathfrak{p}}(3) \quad \text{in } \mathbb{F}_\ell.$$

Since b is invertible modulo ℓ , these are exactly the two compatibility equations with $t = ab^{-1}$. $\square$

Corollary 7.2 (Chebotarev sparsity of common factors). *For a counterexample datum and a sufficiently large good ℓ , every prime $p \equiv 1 \pmod{\ell}$ satisfying the hypotheses of proposition 7.1 lies over the exact compatible Chebotarev set. The ambient set of prime ideals of K_ℓ containing these possible divisors has relative density*

$$\frac{1}{\ell} - \frac{1}{\ell^2} + \frac{1}{\ell^3}$$

in rank three and

$$\frac{1}{\ell} - \frac{1}{\ell^3} + \frac{1}{\ell^4}$$

in rank four.

This is the cleanest connection found between archimedean approximation and the Kummer detector. It does not assign a residue exponent to x ; rather, a prime that divides both approximation errors manufactures the exponent $t = a/b$ modulo ℓ .

7.3 Why density alone is insufficient

Chebotarev describes primes in the ambient set, not prime divisors of one thin sequence. A set of density $1/\ell$ can contain every prime divisor of all the integers $\gcd(E_{a,b}, F_{a,b})$ without contradiction. Neither the prime number theorem nor effective Chebotarev supplies a “generic prime factor” of a prescribed integer.

Likewise, the *abc* conjecture applied separately to

$$m^b = 2^a + E_{a,b}, \quad n^b = 3^a + F_{a,b}$$

would force substantial radical in each error term, but would not force the two radicals to overlap or to escape a specified Frobenius set. Known upper bounds for greatest common divisors of fixed-base sequences, such as the Bugeaud–Corvaja–Zannier theorem, point in the opposite direction: they make large common divisors exceptional. Here a proof needs either a lower bound or a prime-factor distribution theorem for two correlated, nonstationary exponential differences.

7.4 A precise sparse-support target

The following target is intentionally stated at the strength actually needed, not as an informal appeal to “random prime factors.”

Target lemma 7.3 (Frobenius escape for simultaneous near powers). *Let $m, n \in \mathbb{Z}_{>1}$ and let*

$$x = \log_2 m = \log_3 n \notin \mathbb{Q}.$$

Then there exists a sufficiently large odd prime ℓ and infinitely many continued-fraction convergents a/b to x , with $\ell \nmid b$, such that $\gcd(E_{a,b}, F_{a,b})$ has a prime divisor $p \equiv 1 \pmod{\ell}$ whose Frobenius vector is not in $\mathcal{C}_\ell$.

Proposition 7.4. *target lemma 7.3 implies the Alaoglu–Erdős conjecture.*

Proof. Under a counterexample datum, the prime supplied by the target would satisfy the hypotheses of proposition 7.1, which says its Frobenius vector is in $\mathcal{C}_\ell$. This contradicts the conclusion of the target. $\square$

The target may be stronger than necessary, because the gcd can often be one. A more flexible version would replace an actual common divisor by an adelic linear combination of $E_{a,b}$ and $F_{a,b}$ whose product formula forces Frobenius escape. No such construction is presently available. Still, target lemma 7.3 gives a concrete interface for analytic number theory: it asks for a prime divisor in a named conjugacy subset of an explicit pair of integers.

8 The one-more-base route

A different way to break the 2×2 barrier is to manufacture one more multiplicatively independent algebraic base whose x th power is algebraic. The Six Exponentials Theorem then finishes the problem immediately.

Definition 8.1. For a fixed real x , define the algebraic-power domain

$$\mathcal{D}_x = \{a \in \overline{\mathbb{Q}}_{>0}^\times : a^x \in \overline{\mathbb{Q}}^\times\},$$

using the positive real value when $a > 0$.

Theorem 8.2 (third-base exclusion). *Let (x, m, n) be a counterexample datum. If $a \in \overline{\mathbb{Q}}_{>0}^\times$ and a^x is algebraic, then the three logarithms $\log 2, \log 3, \log a$ are linearly dependent over $\mathbb{Q}$. In particular,*

$$\mathcal{D}_x \cap \mathbb{Z}_{>0} \subseteq \{2^r 3^s : r, s \in \mathbb{Z}_{\geq 0}\}. \quad (22)$$

Proof. If the three logarithms were $\mathbb{Q}$ -linearly independent, use them as the three first factors in the Six Exponentials Theorem and use the two $\mathbb{Q}$ -linearly independent numbers $1, x$ as the second factors. The six exponentials would be

$$2, 3, a, 2^x = m, 3^x = n, a^x,$$

all algebraic, contradicting that theorem. For an integer a , a multiplicative dependence among $2, 3, a$ means that no prime outside $\{2, 3\}$ divides a , which proves (22). $\square$

Corollary 8.3 (failure of immediate iteration). *For a counterexample datum, at least one of $m^x = 2^{x^2}$ and $n^x = 3^{x^2}$ is transcendental.*

Proof. By corollary 2.5, at least one of m, n has a prime divisor outside $\{2, 3\}$. If its x th power were algebraic, it would violate theorem 8.2. $\square$

Thus the most obvious attempt to iterate the map $a \mapsto a^x$ is blocked in at least one coordinate. Conversely, this makes the proof target extremely sharp.

Target lemma 8.4 (one-more-base lemma). *Assume $2^x, 3^x \in \mathbb{Z}$ for a real x . Construct an integer a with a prime divisor outside $\{2, 3\}$ such that a^x is algebraic.*

Proposition 8.5. *target lemma 8.4 implies conjecture 1.1.*

Proof. If x were nonintegral, it would be irrational and the constructed a would contradict theorem 8.2. $\square$

The difficulty is that exponentiation respects multiplication but not addition, factor extraction, or polynomial operations. From $m = 2^x, n = 3^x$ one obtains $(2^r 3^s)^x = m^r n^s$, but these bases remain in the rank-two group generated by 2 and 3. A valid third-base argument must supply a new multiplicative direction without using a false identity such as $(m + n)^x = m^x + n^x$.

9 Three proof bridges and one no-go theorem

The preceding sections turn several broad ideas into exact interfaces. This section collects them in a form suitable for a research program or a proof assistant backlog.

9.1 Bridge I: universal local compatibility

The strongest possible local bridge would say that the real exponent relation forces compatibility at every prime. One good Kummer level would already be enough.

Target lemma 9.1 (universal Kummer bridge). *For some odd prime ℓ outside the saturation exceptional set, prove from*

$$\log_2 m = \log_3 n$$

that every unramified Kummer Frobenius vector in W_ℓ belongs to $\mathcal{C}_\ell$.

Proposition 9.2. *target lemma 9.1 implies the Alaoglu–Erdős conjecture.*

Proof. If a nonintegral counterexample had rank four, the compatibility density would be the proper fraction in (8); if it had rank three, it would be the proper fraction in (12). Chebotarev guarantees Frobenius elements outside $\mathcal{C}_\ell$ in either case, contradicting the target. Equivalently, a linear subspace of 2×2 matrices of dimension at least three cannot consist entirely of singular matrices, while $\mathcal{C}_\ell$ is contained in the singular cone. $\square$

theorem 5.2 shows why a continuity proof of this target cannot work: at most primes the normalized orbit has full residue entropy even on a short real arc. Any proof of the universal bridge would therefore have to use a global arithmetic invariant not visible in fixed-prime reduction.

9.2 Bridge II: selected-prime Frobenius escape

This is target lemma 7.3. It asks varying prime divisors of the simultaneous near-power differences to escape a Chebotarev set of relative density about $1/\ell$. It is the most concrete interface with analytic number theory, sieve theory, and arithmetic dynamics.

A potentially more attainable variant is the following product-formula version.

Target lemma 9.3 (adelic escape). *For a counterexample datum, construct from infinitely many convergents a/b a nonzero rational or algebraic number $G_{a,b}$ satisfying all three properties:*

1. *its archimedean absolute value is bounded above by $\exp(-c h(G_{a,b}))$ for some fixed $c > 0$;*
2. *at primes in the compatible Kummer set, its negative valuations have a total contribution $o(h(G_{a,b}))$;*
3. *at incompatible primes, proposition 7.1 or a replacement forces nonnegative valuation.*

Then the product formula gives a contradiction for large height.

This target makes explicit what a vague “combine the real and p -adic estimates” must actually deliver. The hard part is the second condition: Chebotarev density by itself does not control the valuation mass of a thin integer sequence.

9.3 Bridge III: one more algebraic power

This is target lemma 8.4. It is algebraically the cleanest route: one new multiplicative direction invokes Six Exponentials and ends the problem. Candidate constructions should be screened by the following closure test.

Proposition 9.4 (closure test). *Let $G = \langle 2, 3 \rangle \subset \mathbb{Q}_{\geq 0}^\times$. Every base obtained from 2 and 3 using multiplication, inversion, and rational powers remains in $G \otimes \mathbb{Q}$ and cannot serve as the third base. Every base obtained from m, n by the same operations has a known $1/x$ th power, not in general a known x th power. Therefore a successful construction must either make the x -power of an output-side element algebraic or use an operation beyond the multiplicative group while retaining exact control of exponentiation.*

Proof. The first statement is closure of a divisible hull. For the second, if $a = m^r n^s$, then $a^{1/x} = 2^r 3^s$ is algebraic, whereas $a^x = m^{rx} n^{sx}$ contains the unknown quantities m^x, n^x . No multiplicative manipulation removes them unless an additional relation is proved. $\square$

9.4 No-go theorem: exact high directional contact

Theorem 6.1 closes a broad family of auxiliary attacks. It says that a target of the form

$$\text{ord}_0 P(2^t, 3^t) \gg \deg P$$

with algebraic polynomial coefficients is not merely difficult; it is false. A revised auxiliary method must use approximate smallness, multiple points, finite-place divisibility, or a nonpolynomial arithmetic object.

9.5 Architecture scorecard

Architecture	Rigorous gain in this report	Exact missing state-	ment	Main failure mode to avoid
Kummer / Chebotarev	Exact compatible and determinant densities; rank-three transversality	Universal bridge or compatible-prime mechanism	Kummer or a selected	Treating a real exponent as an element of $\mathbb{F}_\ell$ or $\mathbb{Z}_p$
Mantissa dynamics	Conditional real-finite equidistribution and full local entropy	A height-sensitive restriction coupling many places		Inferring finite continuity from a short real arc
Near-power gcd	Every common divisor is localized to the compatible set	Frobenius escape for prime divisors of correlated errors		Applying ambient prime density to a thin sequence
Auxiliary polynomial	Exact equality of directional order and ordinary multiplicity	An approximate adelic replacement with a nonzero lower bound	or	Counting D^2 coefficients as D^2 directional zeros
Six Exponentials	Third-base exclusion and an immediate finishing criterion	Produce one new base a with a^x algebraic		Reusing a base in the divisible hull of $\langle 2, 3 \rangle$

Architecture	Rigorous gain in this report	Exact missing state-	ment	Main failure mode to avoid
Rank-three structure	Möbius normal form and transverse relation determinant	Rule out the remaining fractional-linear case		Assuming transcendence of α implies algebraic independence of arbitrary functions of α

10 Finite verification and reproducible experiments

The density theorems are conceptual consequences of Kummer theory and Chebotarev, but their finite combinatorics should be exhaustively checked before formalization. The following table gives the exact cardinalities for small odd primes.

ℓ	$ \mathcal{C}_\ell $ in rank 4	transverse rank-3 compatible	rank-4 $\Delta = 0$	rank-3 $\Delta = 0$
3	25	7	33	9
5	121	21	145	25
7	337	43	385	49
11	1321	111	1441	121

The ambient cardinalities are respectively ℓ^4 for rank four and ℓ^3 for rank three.

10.1 Exhaustive checker

Listing 1: Exhaustive verification of the compatibility formulas

```

from itertools import product

def compatible(q: int, z: tuple[int, int, int, int]) -> bool:
    u1, u2, v1, v2 = z
    return any(
        (v1 - t * u1) % q == 0 and (v2 - t * u2) % q == 0
        for t in range(q)
    )

def det_zero(q: int, z: tuple[int, int, int, int]) -> bool:
    u1, u2, v1, v2 = z
    return (u1 * v2 - u2 * v1) % q == 0

def in_hyperplane(q: int, rho, z) -> bool:
    return sum(a * b for a, b in zip(rho, z)) % q == 0

def verify(q: int) -> None:
    all4 = list(product(range(q), repeat=4))
    comp = [z for z in all4 if compatible(q, z)]
    assert len(comp) == q**3 - q + 1

```

```

singular = [z for z in all4 if det_zero(q, z)]
assert len(singular) == q**3 + q**2 - q

# Check every nonzero rank-three relation vector rho.
for rho in all4[1:]:
    W = [z for z in all4 if in_hyperplane(q, rho, z)]
    Wc = [z for z in W if compatible(q, z)]
    A, B, C, D = rho
    transverse = (A * D - B * C) % q != 0
    expected = q**2 - q + 1 if transverse else None
    if transverse:
        assert len(Wc) == expected
        assert sum(det_zero(q, z) for z in W) == q**2

for q in (3, 5, 7, 11):
    verify(q)
print("all finite checks passed")

```

The code deliberately defines exact compatibility first and determinant vanishing second. That ordering prevents the false zero-source equivalence in proof-audit warning 3.4.

10.2 Arithmetic experiments suggested by the theory

There is no counterexample to sample. Nevertheless, one can probe the interfaces on arbitrary multiplicatively ranked tuples $(2, 3, m, n)$ and on near-relations. The most informative experiments are:

1. **Kummer histogram.** For fixed ℓ , factor rational primes $p \equiv 1 \pmod{\ell}$, compute order- ℓ residue-symbol coordinates, and compare compatibility and determinant frequencies with the formulas for the observed multiplicative rank.
2. **Rank-three transversality stress test.** Generate tuples with a known primitive relation, separate the cases $AD - BC = 0$ and $AD - BC \neq 0$, and observe the two hyperplane counts in proposition 4.3. The proportional case should always correspond to a rational scalar in the archimedean model.
3. **Mantissa entropy.** For an arbitrary irrational $x = \log_2 m$ (without requiring 3^x integral), compare the empirical symbol distribution of U_j with the exact linear image of (j, k_j) . For a synthetic second coordinate, the determinant controls the transition from a line to the full plane.
4. **Near-power prime factors.** For pairs (m, n) making the two log ratios unusually close, factor $E_{a,b}, F_{a,b}$ along convergents and record Frobenius classes of common prime factors. This does not prove escape, but it can falsify overly strong versions of target lemma 7.3 and identify realistic hypotheses.
5. **Auxiliary contact audit.** Randomly generate algebraic polynomials with prescribed multiplicity at an arithmetic orbit point and verify symbolically that the restricted order equals that multiplicity. Any higher observed order signals a bug in expansion or an accidental zero polynomial.

For large Kummer computations, SageMath or PARI/GP is preferable to a custom discrete-log implementation. The finite count checker above is independent of number-field software and should be included in continuous integration.

11 Lean formalization plan

A productive formalization should not begin with Chebotarev. The results separate into layers with sharply different library requirements.

11.1 Layer 0: finite combinatorics

This layer needs only finite fields and cardinalities.

1. Define

$$\text{Compatible}(u, v) \iff \exists t, v = t u$$

for $u, v \in (\text{ZMod } \ell)^2$.

2. Prove $|\mathcal{C}_\ell| = \ell^3 - \ell + 1$ by splitting on $u = 0$.
3. Define fixed- t planes and prove their pairwise intersection is the zero vector.
4. Prove the transverse hyperplane count $\ell^2 - \ell + 1$ by showing $L + tM$ is nonzero for every t .
5. Prove the singular-matrix counts separately and derive the warning that $\Delta = 0$ is not exact compatibility.

A schematic definition, not intended as drop-in syntax, is:

Listing 2: Lean-shaped statement of exact compatibility

```
def Compatible (u v : (ZMod e11) x (ZMod e11)) : Prop :=
  exists t : ZMod e11, v.1 = t * u.1 /\ v.2 = t * u.2

theorem card_compatible [Fact (Nat.Prime e11)] :
  Fintype.card {z // Compatible z.1 z.2} = e11^3 - e11 + 1
```

11.2 Layer 1: valuation lattices and rank

Represent a finitely supported positive rational by its prime-valuation vector. Formalize:

- rational exponents giving integers are integral;
- the multiplicative-rank floor in proposition 2.4;
- saturation of a finitely generated sublattice and the finite exceptional index I_Γ ;
- the primitive relation in rank three and the determinant argument in lemma 2.6.

The only transcendence input in this layer is that $\log_2 3$ is transcendental. Until Gelfond–Schneider is available in the library, package this as a clearly named imported theorem rather than an opaque axiom local to the proof.

11.3 Layer 2: irrational rotation and mantissa entropy

Formalize theorem 5.1 by reducing it to uniform distribution of $h \mapsto \{c + hx\}$. The proof uses only affine transformations of an irrational rotation and interval measures. Then formalize the linear push-forward in theorem 5.2. This layer is independent of algebraic number theory if $\kappa(2), \kappa(3), \kappa(m), \kappa(n)$ are abstract elements of a finite field.

This separation is valuable: the full entropy theorem can be machine-checked as a statement about an arbitrary 2×2 matrix over $\mathbb{F}_\ell$ before residue symbols are introduced.

11.4 Layer 3: Kummer theory and Chebotarev

Required ingredients are:

- cyclotomic fields containing μ_ℓ ;
- the Kummer isomorphism for a finite-dimensional subgroup of $K^\times/K^{\times\ell}$;
- Frobenius elements and power-residue symbols;
- Chebotarev density for finite Galois extensions;
- the elementary injection of rational Kummer classes into $\mathbb{Q}(\zeta_\ell)$ from lemma 3.1.

If the library lacks a sufficiently general Chebotarev theorem, a sensible intermediate artifact is a theorem parameterized by an abstract “Frobenius-uniformity” hypothesis. The finite formulas and every later logical implication can then be formalized immediately; replacing the hypothesis by Chebotarev becomes an isolated library project.

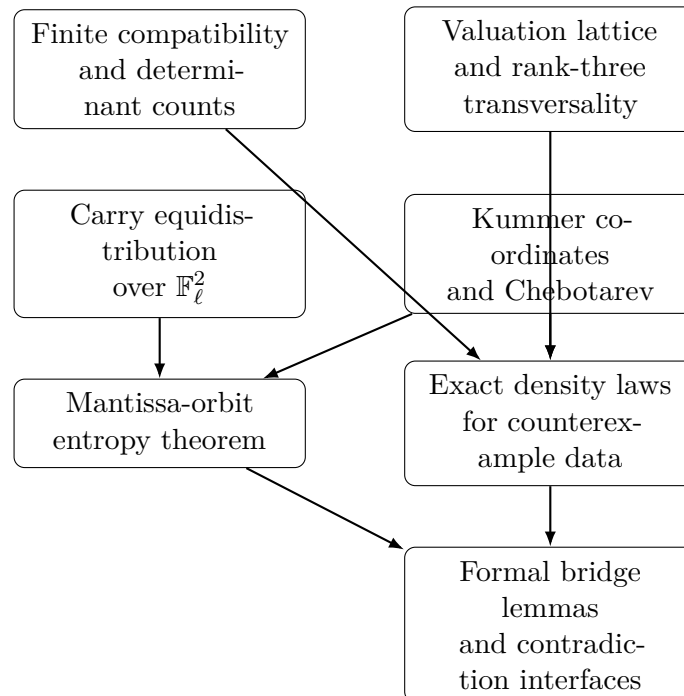
11.5 Layer 4: analytic local rigidity

The proof of theorem 6.1 requires multivariate Taylor expansion, order of a holomorphic zero, and transcendence of α over $\overline{\mathbb{Q}}$. A purely formal-power-series version may be easier: expand

$$e^{t \log 2} - 1, \quad e^{t \log 3} - 1$$

and identify the lowest homogeneous Taylor term of P . Only its linear terms are needed to prove the leading coefficient nonzero. This avoids a large analytic dependency footprint.

11.6 Recommended theorem order



The finite and dynamical branches can proceed in parallel. They provide useful checked artifacts even before the deepest algebraic-number-theory layer is complete.

12 Research agenda after this report

12.1 Highest-priority questions

1. Can common-divisor localization be upgraded to Frobenius escape? The exact proposition is now available. The next step is to search the literature on prime divisors of pairs of exponential-polynomial sequences, especially results that prescribe a conjugacy class rather than merely a congruence class. A useful theorem must be uniform along convergents with both the base and exponent normalization present.

2. Is there an adelic determinant whose forced divisibility beats its archimedean size? The auxiliary-polynomial rigidity theorem says not to seek high exact contact at one point. A determinant involving several near-collision points may still work if obvious powers of $2, 3, m, n$ can be divided out while leaving a nonzero algebraic integer. Every proposed normalization should be audited place by place; dividing by a large real factor without preserving integrality is the standard hidden error.

3. Can one output be made to re-enter $\mathcal{D}_x$? By corollary 8.3, a counterexample forces at least one immediate second iterate to be transcendental. A recurrence, algebraic correspondence, or trace/norm construction that makes either m^x or n^x algebraic would finish in many cases. The construction must exploit integrality in a way not preserved by arbitrary algebraic outputs, or else it would solve the full Four Exponentials problem at once.

4. Can the rank-three Möbius form be attacked separately? In rank three,

$$x = -\frac{A + B\alpha}{C + D\alpha}, \quad AD - BC \neq 0.$$

This is a narrower special-function problem than unrestricted rank four. Possible tools include transcendence measures for fractional-linear functions of logarithm ratios, mixed real/ p -adic regulators, and Hermite–Padé approximation constrained by the integer relation $2^A 3^B m^C n^D = 1$. Merely citing the transcendence of α is insufficient: a rational function of a transcendental number can certainly have algebraic exponentials.

12.2 Directions now deprioritized

The following directions are not impossible in principle, but the results above show that their naive versions cannot be the missing proof.

- **A common p -adic exponent at an arbitrary prime.** Generic Kummer primes are incompatible with probability tending to one.
- **Finite continuity from the real curve.** Conditional mantissa entropy is full at generic primes on every real subarc.
- **Order- D^2 contact from a degree- D algebraic polynomial.** The exact local zero theorem bounds contact by $O(D)$.
- **Chebotarev density as a statement about prime factors.** Ambient density does not imply escape for a thin sequence.
- **A third base built only by multiplying known bases.** This never leaves the rank-two divisible hull.

12.3 A focused next experiment

The most informative computational project is a two-level Frobenius audit. Choose many integer pairs (m, n) for which $\log_2 m$ and $\log_3 n$ agree to unusually high precision but are certifiably unequal. For convergents a/b to their mean, factor the common part of $m^b - 2^a$ and $n^b - 3^a$. At several Kummer levels ℓ , record:

1. whether common factors lie in the exact compatible set as predicted;
2. how much logarithmic mass is carried by each Frobenius class;
3. whether the mass resembles ambient Chebotarev density or is strongly biased;
4. and whether bias correlates with multiplicative rank three versus four.

The experiment cannot establish the conjecture, but it directly tests the only bridge in this report that converts real approximants into selected finite primes. It should also expose whether target lemma 7.3 is realistic or needs an adelic reformulation.

13 Proof-status ledger and conclusion

Statement	Status	Dependencies / caveat
Rational x with $2^x \in \mathbb{Z}$ is integral	proved	Unique factorization
Nonintegral counterexample has transcendental x	proved	Gelfond–Schneider
$\alpha = \log_2 3$ is transcendental	proved	Gelfond–Schneider
Multiplicative rank is at least three	proved	Transcendence of α
Rank-three relation is transverse	proved	Rational case plus primitive relation
Eventual equality of ordinary and Kummer rank	proved	Valuation lattices and cyclotomic ramification
Universal compatibility-density formula	proved	Kummer theory and Chebotarev
Rank-four exact density	proved	Finite counting
Rank-three exact density	proved	Transversality and finite counting
Projective determinant densities	proved	Finite matrix counts
Conditional carry equidistribution	proved	Irrational rotation
Mantissa-orbit entropy theorem	proved	Carry equidistribution and character multiplicativity
Directional order equals ordinary multiplicity	proved	Transcendence of α and Taylor expansion
Common-divisor localization	proved	Residue-character congruences
Third-base exclusion	proved	Six Exponentials Theorem
Universal Kummer bridge	open target	Real relation gives no known finite exponent
Frobenius escape for near powers	open target	Requires prime-factor distribution in a thin pair
One-more-base lemma	open target	Must create a new multiplicative direction
Alaoglu–Erdős conjecture	unresolved here	Any one of the three bridges would finish it

The central lesson is not that local methods fail, but that their correct shape is now visible. A hypothetical counterexample is maximally hostile to a common exponent at generic finite places: exact compatibility has density about $1/\ell$, and the normalized orbit has full two-dimensional residue entropy at almost all Kummer primes. This rules out an entire family of unjustified archimedean-to- p -adic transfers.

At the same time, simultaneous near-power differences provide a legitimate selected-prime bridge, and Six Exponentials turns any third algebraic-power base into an immediate proof. The auxiliary-polynomial theorem eliminates a false parameter gain and redirects interpolation work toward approximate, multipoint, or adelic constructions. These are substantive reductions of the search space even though the final bridge remains open.

The strongest concrete next move is therefore not another broad survey. It is to attack the Frobenius-escape interface for

$$\gcd(m^b - 2^a, n^b - 3^a)$$

along convergents, while formalizing the finite counts and mantissa entropy in parallel. A successful result there would connect the one place where the real exponent genuinely manufactures a finite residue exponent to the place where Chebotarev says such exponents are rare.

A A compact derivation of the rank-three compatibility formula

Let $W = \ker \rho \subset \mathbb{F}_\ell^4$ with $\rho = (A, B, C, D)$ and $AD - BC \neq 0$. For each t , the fixed compatibility plane is

$$\mathcal{C}_t = \{(u, tu) : u \in \mathbb{F}_\ell^2\}.$$

The equation for $W \cap \mathcal{C}_t$ is

$$(A + tC)u_1 + (B + tD)u_2 = 0.$$

The coefficient pair cannot vanish, because that would imply $(A, B) = -t(C, D)$ and hence $AD - BC = 0$. Thus the intersection is a line with ℓ points. Distinct t give lines meeting only at the origin. Hence

$$\left| W \cap \bigcup_t \mathcal{C}_t \right| = 1 + \ell(\ell - 1) = \ell^2 - \ell + 1.$$

Dividing by $|W| = \ell^3$ gives $\ell^{-1} - \ell^{-2} + \ell^{-3}$. This argument is short enough to be the preferred Lean proof.

B Saturation index from a prime-exponent matrix

Let $S = \{p_1, \dots, p_s\}$ contain the prime support of $6mn$. Form the $s \times 4$ integer matrix

$$E = (v_{p_i}(a_j)), \quad (a_1, a_2, a_3, a_4) = (2, 3, m, n).$$

Its column lattice is $L \subset \mathbb{Z}^s$. A Smith normal form

$$UEV = \text{diag}(d_1, \dots, d_r, 0, \dots, 0), \quad d_1 \mid d_2 \mid \dots \mid d_r,$$

computes both the multiplicative rank r and the saturation index

$$I_\Gamma = d_1 \cdots d_r.$$

For every prime $\ell \nmid I_\Gamma$, the reduction of the column span modulo ℓ has dimension r . This supplies an effective list of the Kummer levels excluded by lemma 3.1. In rank three, also exclude prime divisors of $AD - BC$.

C Bibliographic notes

The original conjectural context is the work of Alaoglu and Erdős on highly composite and related integers. The transcendence baseline is the Gelfond–Schneider theorem and the Six Exponentials Theorem. Kummer theory, cyclotomic fields, and Chebotarev are used only in their standard finite abelian form. Uniform distribution enters through the elementary irrational rotation theorem. The selected-prime discussion is informed by the modern literature on gcds of exponential sequences, but no such theorem is invoked in a proof above.

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